

JEE MAIN REVISION MASTER: SEQUENCES AND SERIES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 13 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Arithmetic & Geometric Progressions (AP, GP) & Special Series	13 PYQs	Progressions

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Arithmetic & Geometric Progressions (AP, GP) & Special Series	$S_{\infty} = \frac{a}{1-r} \quad (r < 1), \quad \text{AM} \geq \text{GM} \implies \frac{a+b}{2} \geq \sqrt{ab}, \quad \sum_{n=1}^{\infty} n = \frac{n(n+1)}{2}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Arithmetic & Geometric Progressions (AP, GP) & Special Series (High Yield — 13 PYQs)

Core Principle: A.P.: $T_n = a + (n-1)d$, $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$. G.P.: $T_n = ar^{n-1}$, $S_n = \frac{a(1-r^n)}{1-r}$. Sum of infinite geometric series: $S_{\infty} = \frac{a}{1-r}$ (valid for $|r| < 1$). Sum of first n natural numbers $\sum n = \frac{n(n+1)}{2}$; sum of squares $\sum n^2 = \frac{n(n+1)(2n+1)}{6}$; sum of cubes $\sum n^3 = \left[\frac{n(n+1)}{2}\right]^2$.

■ **Caution / JEE Pitfall:** Infinite GP sum formula $S_{\infty} = \frac{a}{1-r}$ is valid ONLY when common ratio $|r| < 1$.

■ **Speed Trick:** For any three positive numbers a, b, c : $\text{AM} \geq \text{GM}$. Use this inequality to find minimum/maximum values in 10 seconds.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "If we insert two numbers between $\sqrt{2}$ and 4 so that the resulting sequence is in G.P, then the inserted numbers in the order are..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "If 4th, 10th and 16th terms of a G.P. are x, y and z respectively, then..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "If $S_n(n)$ stands for sum to n -terms of a GP with a as the first term and r as the common ratio, then $S_n(n):S_{2n}$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If we insert two numbers between $\sqrt{2}$ and 4 so that the resulting sequence is in G.P, then the inserted numbers in the order are

- (A) $8, \sqrt{2}$
- (B) $2, \sqrt{8}$
- (C) $\sqrt{8}, 2$
- (D) $\sqrt{2}, 8$

Correct Answer: None

Solution Analysis: The sequence after inserting two numbers will be $\sqrt{2}, x, y, 4$. Since these four terms are in G.P., let the common ratio be r . The first term is $\sqrt{2}$ and the fourth term is 4. For a G.P., the fourth term is ar^3 . So, $\sqrt{2}r^3 = 4$
 $r^3 = \frac{4}{\sqrt{2}}$ Now, $\frac{4}{\sqrt{2}} = \frac{2^2 \cdot 2^{1/2}}{2^{1/2}} = 2^{3/2}$ So, $r^3 = 2^{3/2}$ Taking cube root, $r = 2^{1/2} = \sqrt{2}$ Now the terms are:
 $\sqrt{2}, \sqrt{2} \times \sqrt{2}, 2 \times \sqrt{2}, 4$ So, $x=2$ and $y=\sqrt{8}$ Hence, the inserted numbers in order are $2, \sqrt{8}$ Correct option: Option B.